

APPLETON, MN, USA

WMO: 726466

Lat: 45.228N

Lon: 96.007W

Elev: 311

StdP: 97.65

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54917

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.3	(c) -23.1	(d) -29.1	(e) 0.3	(f) -25.5	(g) -27.2	(h) 0.3	(i) -23.4	(j) 14.6	(k) -9.9	(l) 12.9	(m) -11.0	(n) 2.7	(o) 250	(p) 0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	32.4	23.0	31.0	22.5	28.9	21.2	25.4	30.0	24.2	28.4	23.1	27.2	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.5	28.1	22.7	18.0	27.0	22.1	17.4	26.4	79.8	29.7	74.2	28.3	70.1	27.4	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	10.9	9.3	-30.0	35.2	3.1	1.8	-32.2	36.4	-34.0	37.4	-35.7	38.4	-38.0	39.7
			-30.3	27.3	3.0	1.5	-32.5	28.4	-34.3	29.2	-36.0	30.0	-38.2	31.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-10.5	-8.7	-1.4	6.9	14.0	19.7	22.1	20.3	16.3	8.3	0.2	-7.6
	DBStd	12.83	7.43	7.37	7.33	5.72	5.06	3.65	3.35	3.36	4.76	5.40	6.17	6.93
	HDD10.0	2606	634	524	358	123	20	0	0	0	6	98	297	544
	HDD18.3	4565	892	758	612	344	152	27	8	18	93	315	543	803
	CDD10.0	1409	0	0	5	31	144	290	376	321	195	44	4	0
	CDD18.3	327	0	0	0	1	18	67	126	80	33	2	0	0
	CDH23.3	3279	0	0	2	28	242	667	1219	721	357	43	1	0
	CDH26.7	1057	0	0	0	7	81	203	425	216	114	11	0	0
(14) Wind	WSAvg	4.0	4.2	4.4	4.5	4.8	4.7	3.9	3.1	2.9	3.5	4.1	4.2	4.1
(15) Precipitation	PrecAvg	646	18	18	34	64	72	98	96	73	71	61	27	16
	PrecMax	821	46	40	89	171	118	191	187	133	158	185	95	49
	PrecMin	416	1	4	8	11	17	36	32	15	12	6	0	1
	PrecStd	101	12	10	20	36	27	45	40	34	37	47	25	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	12.0	22.0	27.2	32.5	33.0	34.1	33.8	32.6	27.9	19.8	10.1
		MCWB	3.3	9.6	13.1	15.6	19.4	21.9	25.4	24.4	21.5	16.7	12.7	6.0
	2%	DB	2.9	6.3	16.1	22.7	28.8	31.2	32.4	31.1	29.9	23.0	15.8	6.1
		MCWB	1.1	3.9	11.2	12.9	17.4	21.3	24.5	23.7	19.9	14.4	10.1	2.8
	5%	DB	2.1	3.0	12.3	19.8	26.1	28.8	31.0	28.8	27.5	20.2	12.3	2.9
		MCWB	0.9	1.7	8.9	11.6	16.4	20.3	23.8	22.4	19.4	13.6	8.1	1.3
	10%	DB	0.0	2.1	8.0	17.1	22.9	27.4	28.9	27.4	25.0	17.3	8.8	2.0
		MCWB	-0.6	1.1	5.7	10.2	15.1	19.4	22.4	21.2	18.0	11.9	5.6	0.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	8.3	14.8	16.7	21.9	24.9	27.9	26.9	24.0	19.6	14.3	6.8
		MCDB	6.6	10.9	17.6	24.7	28.5	30.0	31.6	30.0	29.1	24.7	18.1	9.7
	2%	WB	1.7	4.2	11.7	14.3	19.6	23.5	25.9	24.9	22.5	16.9	10.6	3.1
		MCDB	3.0	6.2	15.4	20.3	25.6	28.4	30.3	29.5	27.7	20.7	14.2	5.2
	5%	WB	0.6	2.3	8.5	12.5	17.9	22.2	24.8	23.5	21.1	14.2	8.1	1.7
		MCDB	1.7	3.4	11.9	18.2	23.0	27.0	29.1	27.4	25.5	18.7	11.9	3.1
	10%	WB	-0.9	0.7	5.8	10.8	16.3	20.7	23.6	22.3	19.5	12.4	5.9	0.3
		MCDB	-0.3	1.8	8.6	15.6	22.0	25.3	27.5	26.2	23.5	16.5	8.9	1.6
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.0	9.8	11.7	12.2	11.5	11.7	11.8	13.0	11.6	10.0	9.0
		MCDBR	11.4	10.9	15.1	17.6	16.4	14.2	12.9	13.6	15.8	16.9	15.2	11.5
	5% WB	MCWBR	9.8	8.7	9.9	9.3	7.7	6.7	6.5	7.0	7.7	9.2	10.0	9.1
		MCDWR	10.6	10.0	14.5	14.9	13.1	11.5	11.4	12.1	13.1	14.1	14.2	10.6
(40) Clear-Sky Solar Irradiance	taub		0.266	0.277	0.308	0.360	0.382	0.391	0.394	0.401	0.363	0.316	0.281	0.262
		taud	2.436	2.418	2.440	2.338	2.342	2.374	2.350	2.315	2.406	2.520	2.530	2.462
	Ebn at Noon		861	918	929	900	887	877	869	849	856	860	840	830
		Edh at Noon	69	86	97	118	122	118	120	120	101	78	64	61
(44) All-Sky Solar Radiation	RadAvg		1.77	2.85	3.80	4.77	5.48	6.12	6.53	5.43	4.19	2.69	1.75	1.28
	RadStd		0.20	0.32	0.35	0.41	0.40	0.39	0.24	0.45	0.30	0.36	0.17	0.14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.95	-0.91	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page